Florida

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1 Introduction

This study examines whether the temperature in Key West, Florida (USA), experienced a significant upward trend during the 20th century. The dataset covers a 100-year period, from 1901 to 2000, and includes annual mean temperature observations.

2 Methods

To evaluate the relationship between time and temperature, Spearman's rank correlation coefficient (rho) was employed. A permutation test was performed by randomly shuffling the year–temperature pairs 10,000 times to create a reference distribution of rho values. The observed rho from the actual data was then compared against this random distribution to determine statistical significance.

3 Results

The computed Spearman's rank correlation coefficient (rho) was 0.526. This statistic exceeded all 10,000 randomly generated rho values, resulting in a p-value of 0. The outcome provides strong evidence that the mean annual temperature in Key West, Florida, rose significantly over the 20th century.